FR. CONCEICAO RODRIGUES COLLEGE OF ENGINEERING

Department of Computer Engineering

Academic Year 2025-26

Subject Name :*BDA Lab*

Class : B.*E. ComputerEngineering*

Semester : VII **Subject Code :CSL702**

Practical No:	4
Title:	Write a program in python to implement Matrix-Vector Multiplication algorithm using Map Reduce
Date of Performance:	07-08-2025
Roll No:	9913
Name of the Student:	Mark Lopes

Evaluation:

Performance Indicator	Below average	Average	Good	Excellent	Marks
On time Submission (2)	Not submitted (0)	Submitted after deadline (1)	Early or on time submission(2)	---	

Test cases and output (4)	Incorrect output (1)	The expected output is verified only a for few test cases (2)	The expected output is Verified for all test cases but is not presentable (3)	Expected output is obtained for all test cases. Presentable and easy to follow (4)	
Coding efficiency (2)	The code is not structured at all (0)	The code is structured but not efficient (1)	The code is Structured and efficient. (2)	-	
Knowledge(2)	Basic concepts not clear (0)	Understood the basic concepts (1)	Could explain the concept with suitable example (1.5)	Could relate the theory with real world application(2)	
Total					

Objective:

- Understand Matrix Vector Multiplication
- Use MapReduce framework forMatrix Vector Multiplication
- Write python program to solve Matrix Vector Multiplication

Input:

Matrix.txt

M 3,2
1,1,3
1,2,4
2,1,5
2,2,6
3,1,7
3,2,8

Vector.txt V,2

1,1

2,1

Output: 11

17

23

Code:

```
# Matrix Vector Multiplication using coordinate format

def read_matrix(file):
    matrix = {}
    rows, cols = 0, 0
    with open(file, "r") as f:
        header = f.readline().strip()
        if header.startswith("M"):
            # Example: "M 3,2"
            dims = header.split()[1]
            rows, cols = map(int, dims.split(","))
        for line in f:
            if line.strip():
                r, c, val = map(int, line.strip().split(","))
                matrix[(r, c)] = val
    return matrix, rows, cols


def read_vector(file):
    vector = {}
    size = 0
    with open(file, "r") as f:
        header = f.readline().strip()
        if header.startswith("V"):
```

```
            # Example: "V,2"
            size = int(header.split(",")[1])
        for line in f:
            if line.strip():
                i, val = map(int, line.strip().split(","))
                vector[i] = val
    return vector, size


def multiply(matrix, rows, cols, vector):
    result = [0] * rows
    for (r, c), val in matrix.items():
        result[r - 1] += val * vector.get(c, 0)
    return result


if __name__ == "__main__":
    matrix, rows, cols = read_matrix("/content/matrix.txt")
    vector, size = read_vector("/content/vector.txt")

    if cols != size:
        print("Matrix column count must equal vector size!")
    else:
        result = multiply(matrix, rows, cols, vector)
        for r in result:
            print(r)
```

I/P:

matrix.txt

```
M 3,2
1,1,5
1,2,6
2,1,8
2,2,9
3,1,11
3,2,12
```

matrix.txt | vector.txt

```
V,2
1,1
2,1
```

O/P:

```python
        print("Matrix column count must equal vector size!")
    else:
        result = multiply(matrix, rows, cols, vector)
        for r in result:
            print(r)
```

```
11
17
23
```